

Project Initialization and Planning Phase

Date	7 July 2026
Team ID	XXXXX
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Problem Statement	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A farm owner or agricultural worker.	Determine the most profitable and suitable crop for my specific soil and weather conditions.	I don't know if my land satisfies the precise chemical and environmental requirements for high-yield crops.	Soil conditions and climate patterns change, and guessing leads to massive financial loss.	Uncertain, stressed, and hesitant to invest in seeds.
PS-2	An agricultural consultant or agronomist .	Quickly evaluate large plots of land and give accurate crop recommendations to my clients.	Manual soil analysis and cross-referencing weather charts takes immense time and effort	Every recommendation requires checking multiple complex environmental factors simultaneously.	Overloaded, inefficient, and concerned about giving inaccurate advice.

PS-3	A land investor	Understand the agricultural potential of newly purchased land instantly.	I have no easy way to estimate crop eligibility without hiring an expensive expert.	Traditional evaluation happens only after weeks of lab testing.	Curious but unsure about my financial risk.
PS-4	A government agricultural department.	Improve the efficiency of distributing crop subsidies based on optimal regional crops.	Manual decision-making may lead to delays and suboptimal crop recommendations for regions.	Human assessment can vary depending on workload and lack of data aggregation.	Concerned about operational efficiency and resource waste.
PS-5	A first-time modern farmer.	Receive guidance on modern crop selection before planting.	I am unfamiliar with the soil factors (N, P, K, pH) that affect crop yield.	I have limited knowledge about soil chemistry and its impact on specific crops.	Confused and lacking confidence in my decisions.